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Stair mobility assistive system

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ABSTRACT
Individuals with mobility limitations encounter daily difficulties navigating stairs, often finding existing systems inconvenient for themselves or their assistants. This paper introduces a new assistive system tailored to make stair navigation easier and more convenient for them. The system features a simple and practical design that can be installed on most staircases without major modifications, enabling quick setups to assist people with temporary or permanent mobility limitations. It utilizes a piggyback-style configuration with two prismatic joints that employ PID velocity control, automatically adjusting to different users or chair types, thus reducing the need for prior knowledge of chair or user payloads. The structure allows users to remain in their own wheelchairs or use standard chairs, reducing the need for chair transfers that typically require assistance, thereby indicating the potential to reduce reliance on external helpers. The system's ease of parameter reconfigurability suggests improved adaptability and practicality, extending its usability to home and healthcare facilities. Performance evaluation results indicate that the system can address key limitations observed in current implementations within the scope of the prototype.

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1. Introduction

In the U.S., 1.3% of the population has impairments or restricted mobility, and this group is not limited to the elderly (Chocoteco et al., 2015). Among wheelchair users, 28% are under the age of 60. Additionally, 10.2 million people (4.4% of the population) use walkers, crutches, or canes. Whether due to permanent or temporary disabilities, these individuals face significant challenges when navigating stairs in places that lack basic assistive options like elevators or escalators. Injuries that affect limbs, sudden health issues like kidney failure requiring dialysis, or other unforeseen circumstances can lead to sudden mobility restrictions, drastically altering people's lifestyles. This is particularly problematic for those living in private homes or residential buildings without elevators or escalators, as they must rely on self-help, family members, or healthcare professionals to navigate stairs. Addressing this issue requires careful planning due to physical and structural challenges in the building or in the staircase. Whether the disability is temporary or permanent, a quick solution is needed to help individuals move safely up and down stairs. However, adding an elevator or escalator might not be practical due to high costs, extensive structural changes, and the need for approvals. Additionally, the geometry of the staircase could impose limitations on such modifications. Therefore, there is a clear need for a practical, cost-effective, adaptable, easily re-scalable and user-friendly stair-climbing assistive system that can overcome the current systems shortcomings which are explained next.

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1.1. Background

Assistive technologies for staircases are readily available in the market, designed to aid individuals with mobility or balance challenges in moving on the stairs safely. These technologies encompass several options, ranging from basic aids like handrails to more sophisticated products such as stair lifts, custom-made wheelchairs, platform lifts, and legged robots. Nonetheless, solutions that provide promising options such as platforms and wheelchairs come with disadvantages, primarily driven by their high costs resulting from intricate designs utilizing numerous sensors and actuators, as well as their considerable weight and bulkiness, limiting their use in confined spaces. Many researchers proposed to handle these cases by the use of ground vehicles-based systems that are capable of dealing with stairs (Pappalettera et al., 2023), negotiating obstacles (Alamdari & Krovi, 2016), or uneven terrain (Wei et al., 2022). They suggested ways to enhance the performance of wheelchairs, which are often classified as stair-climbing robots (Quaglia et al., 2011), with the aim of improving their efficiency in navigating stairs. For instance, a wheelchair equipped for climbing stairs was unveiled in Quaglia and Nisi (2017). It utilized a cam-operated system for motion control. The designers prioritized stability to enhance user comfort during operation. Nevertheless, the design yielded a complex hybrid wheel-leg locomotion mechanism with a triple-wheel cluster configuration. Despite efforts, complete elimination of oscillation proved unattainable, necessitating slower motion to reduce its impact.

One common design strategy in research is wheelchairs that use a leg-wheel locomotion system as in Bang et al. (2011). These designs combine the best aspects of legged mobility on stairs with the efficiency of wheels on level surfaces. Each wheel of the prototypes described in these publications can move up steps employing an articulated system. Although this method works well, it requires a complex control scheme and a large number of motors.

In an effort to substitute human assistance with automation, Ikeda et al. (2021) presented an assistive robot designed to collaborate with a wheelchair in aiding users in ascending stairs. This concept, requiring coordination between paired manipulators and a sensor-equipped wheelchair, yielded a solution that is expensive, cumbersome, and complex. Consequently, sophisticated controllers are necessary, hindering its viability in the market for mobility assistance devices tailored for stair navigation. Moreover, combining two or three different locomotion strategies, such as tracked, wheeled, and legged systems, into hybrid mobility systems is another way to develop stair navigation mechanisms. The mechanism outlined in Chocoteco et al. (2015) combined two different ways of locomotion mechanisms, one that uses wheels and the other that uses sliding supports. Although the system demonstrated a high degree of proficiency in ascending stairs, it encountered difficulty with various stair inclinations, which limits its climbing capacity and could compromise human safety. This drawback limits how widely the system may be used because a setup that works well for one stair geometry might not work well for another. The main drawbacks of wheelchair stair navigators are static stability, especially on stairs with high slopes, then the discomfort due to seat oscillation during the motion. In addition, the user must use a custom-made stair-climbing wheelchair, which can be costly, especially for individuals with temporary injuries.

Another important direction that researchers followed to deal with the problem is stair climbing platforms (Pappalettera et al., 2023). They comprise a carrier-type robot consisting of a platform mechanism, responsible for balancing the payload in a horizontal pose, mounted on a robotic carrier responsible for negotiating the stairs during the ascend or descent motion. The carriers can be track-based carrier types robots (Yoneda et al., 2009), wheel cluster-based carrier types (Onozuka et al., 2020), or hybrid and legged-based types. However, these systems require advanced automation, involving complex robotic designs, resulting in intricate and costly designs (Wen et al., 2019).

Industrial lifting platforms have gained significant attention for their ability to navigate multiple steps (Robert, 2019). However, from a commercial standpoint, these solutions are deemed unfavorable because they detract from the staircase's aesthetics, leading to an unattractive appearance and hindering other users' freedom of movement on the stairs.

Common metrics like payload capacity and speed offer significant indicators of performance. Moreover, addressing challenges related to maximum crossable step heights and stair slopes is crucial, and therefore, these factors can be incorporated into the performance evaluation metrics to assess the effectiveness of a conceptual design in offering a viable solution for stair climbing.

1.2. Contributions and objectives

This work introduces a novel stair-climbing assistive system designed to serve as either a temporary or permanent mobility aid for navigating stairs. The features of this design are derived from the shortcomings of current systems, which serve as benchmarks for evaluation. This study is guided by the following research question: Can a universally adaptable, low-cost stair-mobility assistive system be designed to accommodate standard chairs and wheelchairs while ensuring stability and requiring minimal structural modification?

The main contributions of the proposed system include:

- Universal Chair Adaptability: unlike conventional stair-climbing systems that require specialized chairs or wheelchair designs or customized permanent chairs integrated into the mechanism, the proposed system accommodates standard off-the-shelf chairs and wheelchairs, significantly reducing cost and enhancing applicability.
- Minimal Staircase Modification and Extended Accessibility: The proposed design allows for easy installation on existing staircases with minimal alterations, avoiding permanent structural changes and ensuring that the staircase remains fully accessible when the system is not in use.
- Adaptive Control and User-Centric Reconfigurability: The system employs PID velocity control to automatically adapt to variations in user weight, stair inclination, and dynamic frictional forces, while also allowing user-defined customization of parameters such as lifting speed, user size, and payload capacity to accommodate different chairs or users. This combination ensures personalized mobility solutions, scalability, and seamless integration into diverse environments.
- Improved Stability and Comfort: By employing prismatic joints-based movement, the system minimizes oscillations and ensures a smooth, safe ride compared to traditional track-mounted solutions.
- Cost-Effectiveness and Ease of Use: the simplified mechanism avoids bulky, complex automation, making it a practical, low-cost alternative for both home and healthcare environments.

It is important to note that this study presents a proof-of-concept prototype, and the results are intended to demonstrate feasibility rather than full-scale real-world performance

2. Design

The system utilizes motion in two directions, necessitating the use of two actuators. One actuator is responsible for providing vertical motion through a linear slider, while the other actuator drives a carriage that moves along the stairs using a cable-driven mechanism.

Attached to the bracket of the linear slider is a foldable linkage with 90-degree joint limits. When the mechanism is not in use, the links fold back to their home position against the wall, freeing the entire stair space for other users (Figure 1(a)). When the system is in use, the linkage is positioned horizontally to serve as a platform for mounting either a standard chair or a wheelchair (Figure 1(b)). The linear slider, combined with the foldable linkage, accommodates chairs of varying heights and shapes, including wheelchairs, by allowing them to be mounted on the horizontal links. Consequently, the slider can elevate the chair to a height that prevents the seated person's feet or any parts of the chair from hitting the stairs during the system's traversal movement along the stairs, Figure 1(c).

A key operational advantage of the proposed system is its ability to accommodate a wide range of standard chairs, including wheelchairs. Typically, existing stair-climbing systems require wheelchair users to transfer from their personal wheelchair to a dedicated seat, with additional helpers often needed to assist with the transfer and to carry the wheelchair to the other end of the stairs. Likewise, individuals using crutches or with limited mobility generally need assistance to be seated on the system's dedicated chair. In contrast, the proposed system is designed to accommodate wheelchair users while they remain seated, thereby reducing the need for such transfers in the tested scenarios – a task that can cause

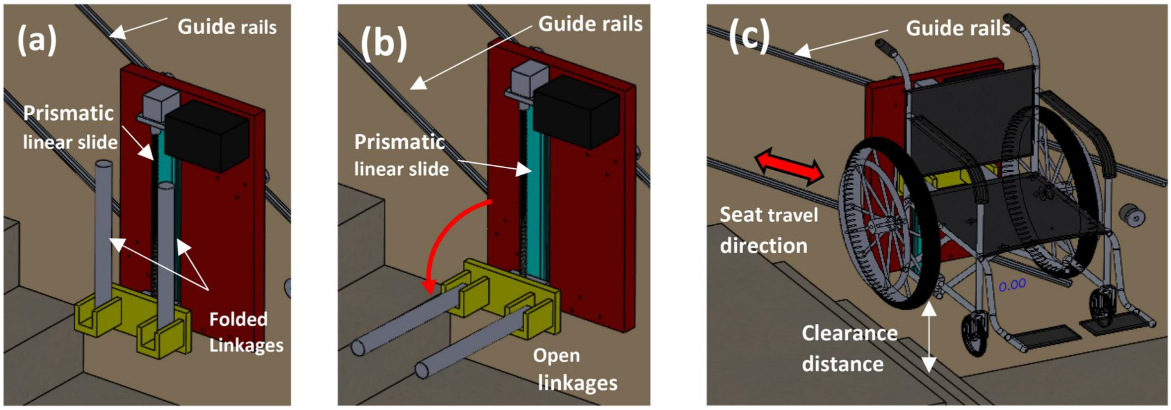

Figure 1. (a) Mechanism folded against the wall to free space when not in use; (b) mechanism ready to accommodate various chair types for stair climbing; (c) vertical linear slider assures lifting the chair and patient above stairs.

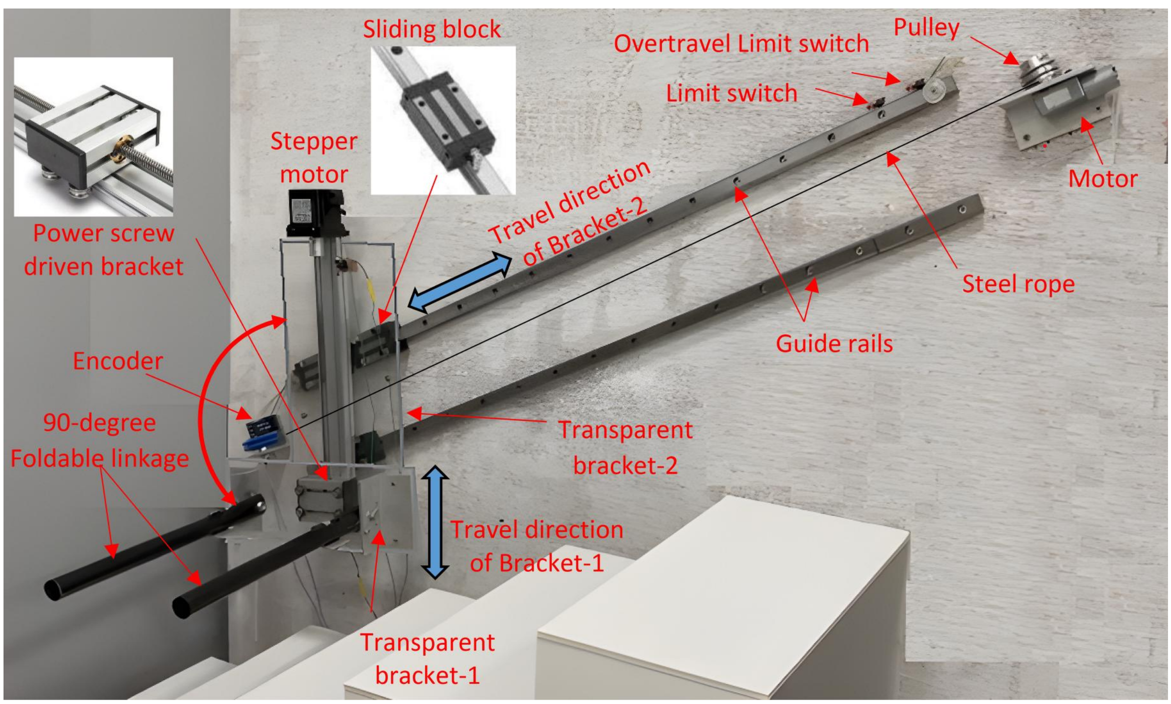

Figure 2. Proposed system components.

strain for caregivers. Furthermore, the ability to accommodate standard chairs eliminates the need for specially customized seating solutions. In subsequent sections, the operating sequence involving a wheelchair is demonstrated in the 3D simulations, where the wheelchair remains positioned on the foldable linkages throughout the operation. In the experimental validation, the system was tested using a standard chair configuration, which users were able to position and utilize without manual lifting or transfers. Consequently, the system indicates the potential to reduce reliance on additional helpers and minimize the need for specialized stair-climbing wheelchairs or bulky, permanently mounted systems.

To construct a proof-of-concept prototype, a systematic design process is adopted to define the requirements and specifications for a full-scale system. The key design criterion is the system's ability to safely handle a total payload that includes both the user's body weight and the wheelchair weight.

A nominal realistic user's payload consists of an 85–100 kg. This payload directly influences the required capacity of the motors and other structural and mechanical components, as discussed later. For experimental validation, a scaled-down prototype supporting a rated payload capacity of up to 5 kg was developed to demonstrate the system's functionality and verify the underlying design principles. Figure 2 illustrates the components of the experimental setup.

A prismatic joint formed by a linear slider made up of a motor-driven power screw is utilized to move carriage bracket-1 through vertical travel that reaches up to a maximum of 350 mm, where this user-defined distance can be chosen to meet the customer's requirements. The lifting force can be customized based on the specifications of the driving motor and power screw to meet customer requirements. Mechanical limit switches are installed at both ends of the lateral motion to define the travel limits along the stairs, while additional redundant switches are included to prevent over-travel. Mechanical limit switches are also used to define the vertical travel stroke of the linear slider. However, an alternative option to define the vertical stroke is by programming start and end points using a microcontroller, providing more flexibility in choosing the desired distance. Once the set end-of-travel point is reached, the motor is locked by an H-bridge brake applied by the microcontroller. This ensures the seat remains at a specific height during ascent or descent on the stairs. Additionally, the motor can be equipped with a normally locked electromagnetic brake that engages when power is off and disengages when power is on.

The lateral motion system, which is responsible for moving the vertical linear slider and the chair along the stairs in both ascending and descending directions, utilizes a cable-driven prismatic joint. A motor is fixed at one end of the stairs, while an idler pulley is positioned at the opposite end. A steel rope or cable is used to drive bracket-2. This transverse bracket-2, responsible for supporting the payload of the chair/user and the linear slider, is outfitted with ball bearings that allow sliding bracket-2 along two guide rails affixed to the wall. The choice of the number of guides and gliding blocks depends on the payload requirements. Note that the brackets mounted on both prismatic joints were made of transparent acrylic glass for better visibility of the system components. It is evident that the main alteration or addition required on the staircase involves attaching the rails to the walls.

Bracket-2 glides along the guide rails using four sliding blocks, with two blocks on each rail. This configuration ensures dynamic stability, preventing the linear slider, and consequently the chair, from rotating. It effectively maintains an upright position for the chair during lateral motion. Additionally, this arrangement helps eliminate any inconvenient oscillatory motion typically associated with wheelchair or track-based systems.

Similar to the linear slider, mechanical limit switches are located at both ends of the lateral travel path to set the end-of-travel limits. Once the seat reaches its designated endpoint of travel, these switches serve to interrupt the motor's operation and stop the motion.

The system is equipped with several safety measures designed to ensure reliable operation under a range of potential failure scenarios. The following section details the identified failure modes and the associated fail-safe countermeasures:

- Power failure and brake deactivation: In the event of a power outage, the H-bridge brakes controlled by the circuit may be deactivated, potentially leading to chair over-speeding. To address this, the system incorporates normally-locked electromagnetic brakes that automatically engage during power loss or disconnection, locking the motor and halting motion. In addition, an emergency stop button allows the user or an assistant to immediately cut power and engage the brakes.
- Speed controller malfunction or cable detachment: A failure in the speed controller or detachment of the cable-driven joint could cause uncontrolled descent. To mitigate this risk, the vertical linear slide is connected to a ratchet-driven cable wrap installed at the top of the stairs. Functioning similarly to an automotive seatbelt ratchet, this mechanism locks when the chair exceeds a predefined speed, preventing dangerous acceleration.
- Over-travel protection: Redundant limit switches are installed as backups to the primary limit switches at both ends of the travel path. These ensure that motion is stopped safely if the main switches fail to detect the end-of-travel position.

3. Methodologies

The modular nature of the design enables scalability by upgrading actuator torque, structural dimensions, and materials to support human loads exceeding 100 kg. Analytical torque and power calculations confirm that increasing motor specifications and using aluminum or steel structural components would

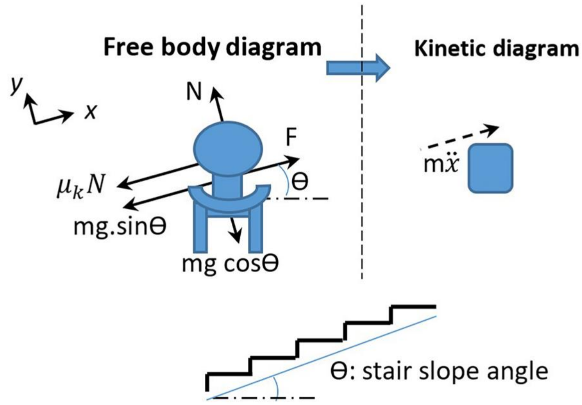

Figure 3. Dynamic model of the system (lateral joint).

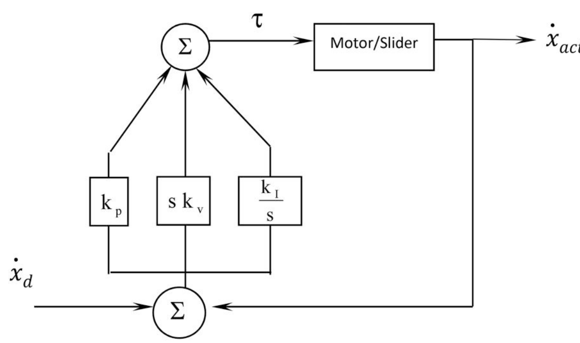

Figure 4. PID velocity control is used to glide the chair along the stairs.

meet these requirements. However, Full-scale human testing is outside the scope of the present study and will be pursued in subsequent research. Therefore, the prototype was tested with payloads up to 5 kg to experimentally verify the system's motion control and mechanical integrity. This scale was selected to demonstrate proof of concept safely and efficiently.

The lateral motion motor is controlled by a PID controller to regulate and limit the chair's sliding speed, ensuring it doesn't exceed a predefined safe limit. The benefits of using a velocity mode controller for the system are as follows: 1) Operating at a slower speed reduces the system's jerking movements caused by inertia when the brakes are suddenly applied after the limit switches at the end of the travel are triggered; and 2) the dynamic model of the lateral slider subsystem, based on the free body diagram in Figure 3, is given by the equation:

$$F = m\ddot{x} + \mu_k N + mgsin(\theta) \tag{1}$$

where F is the total force acting on the system, m represents the total mass, which includes the combined weights of the user, the chair, and the lifting prismatic joint, $\ddot{x}$ is the system's acceleration, θ is the stair slope angle, μ_k is the coefficient of kinetic friction and N is the normal force. It is important to note that the torque experienced by the joint-driving motor depends on both the stair slope angle θ and the total mass m, which varies depending on the combined weights of the chair and user. Thus, when operating the slider bracket in velocity control mode, the controller compensates for gravity without needing prior knowledge of the stair slope angle, user weight, or the chair's type and weight- factors that change with different users. The PID velocity controller is shown in Figure 4.

The used PID gains are given as: kp = 4, kv = 0.25 and ki = 0.5. The lateral motion of the system is affected by the drag force due to unknown dynamic coefficients of friction μ_k in the linear bearings and velocity-dependent viscous damping. To empirically measure this drag force, a steel rope was used to pull bracket-2 at a constant desired speed up the stairs, as illustrated in

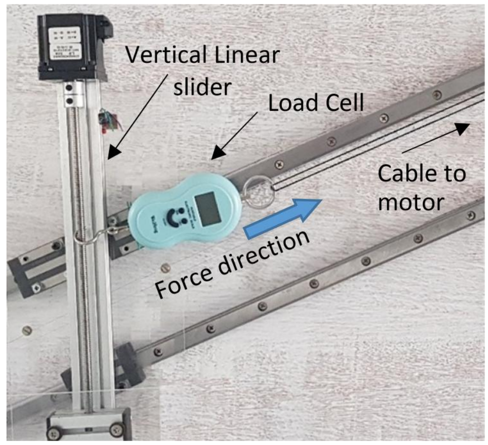

Figure 5. Experimental setup for measuring the force required to drag the user's seat while ascending stairs.

Table 1. Comparison of motor torque and power requirements for 5 kg and 100 kg payloads.

Parameter	Formula	5 kg prototype	100 kg (projected estimate)
Payload		5 kg	100 kg
Force	$F \propto m$	50 N	1000 N
Torque (No SF)	F x r	87.5 N·cm	1750 N.cm
Torque (with SF = 1.3)	$\times$ 1.3	11.8 kg·cm	236 kg·cm (23.15 N·m)
Power (with SF = 1.3)	$F.v \times 1.3$	3.25 W	65 W

Figure 5. A load cell attached to bracket-2 measured the drag force.

The subsequent steps were followed to calculate the motor torque and power.

- Torque Calculation: for safety considerations, a lateral travel speed of 0.05 m/s was selected. The measured drag force was approximately 50 N. With a motor pulley diameter of 3.5 cm (radius of 1.75 cm), the required motor torque was calculated as 50 N × 1.75 cm = 87.5 N·cm, or 8.9 kg·cm.
- Application of Safety Factor: To ensure a sufficient margin, a safety factor of 1.3 was applied, increasing the required motor torque to 8.9 kg·cm × 1.3 = 11.8 kg·cm.
- Power Calculation: The required motor power was then calculated using the formula

$$P = F.v \tag{2}$$

where P is the motor power in watts, F represents the tension force in the cable, and v denotes the lateral travel velocity.

While the proof-of-concept prototype demonstrated a 5 kg payload capacity, the system's modular design allows for customization of the payload capacity through the selection of motors with higher torque and adjustments to the prismatic joint lifting mechanism.

Using the measured force (50 N), the chosen velocity (0.05 m/s), and the safety factor (1.3), the minimum required motor power was calculated as 50 N × 0.05 m/s × 1.3 = 3.25 Watts.

To evaluate the system requirements under realistic operating conditions, the same calculation procedure was applied, considering a payload of 100 kg for the user's weight. A linear relationship between payload and drag force was assumed as a first-order engineering approximation; this assumption has not been experimentally verified at full scale and should be regarded as a preliminary engineering estimate rather than validated performance data. The corresponding motor torque and power requirements were determined using the same pulley dimensions, travel speed, and safety factor adopted for the prototype analysis. Full-scale experimental validation of these projections is planned in future work. The estimated values for both the 5 kg prototype and the 100 kg projected full-scale configuration are summarized in Table 1.

4. Validation, discussion and practical considerations

A step-by-step explanation of the system's operation is provided below, corresponding to the sequence shown in Figure 6. The process is designed to be intuitive for the user. Initially, the foldable linkage is

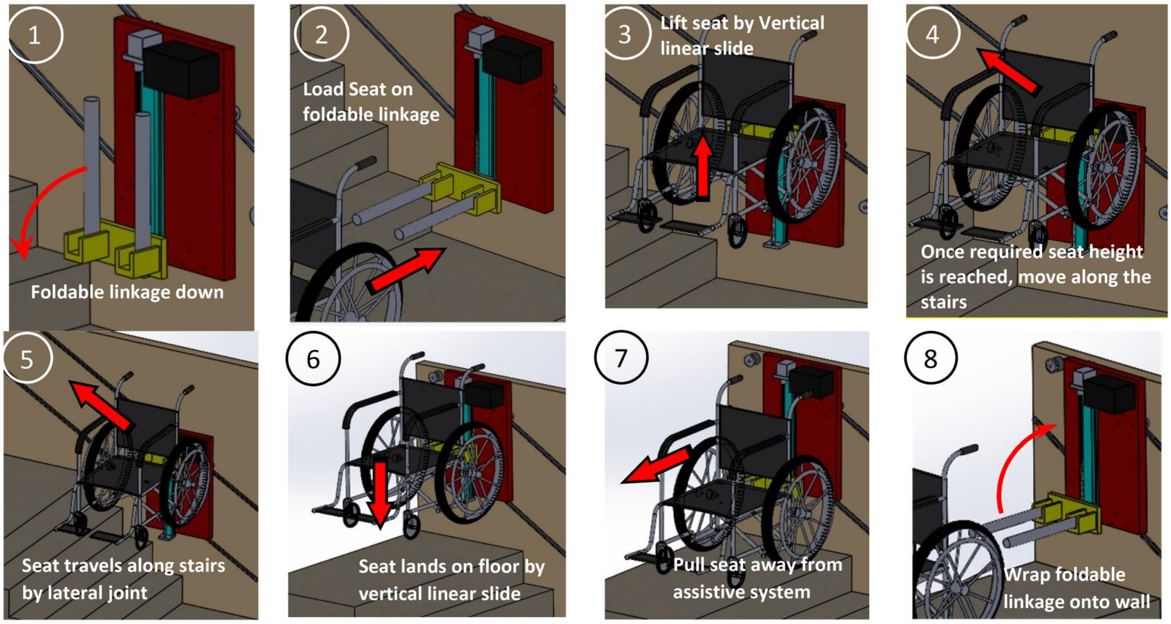

Figure 6. A sequence of operation of the system showing the step-by-step operating scenario.

Figure 7. System testing and validation.

placed in its vertical position, with the vertical linear slide at its lowest level (Figure 6.1). When the system is needed, the linkage is lowered to a horizontal position, and the chair is slid onto the linkage (Figure 6.2). The user then secures themselves using the seatbelt.

After the system is activated, the vertical linear slide lifts the linkage and chair until the user's feet clear the stairs. The motion continues until the upper limit switch is reached, indicating the correct elevation has been achieved (Figure 6.3). Next, the lateral motion motor moves the chair horizontally across the staircase under controlled velocity (Figures 6.4 and 6.5). When the lateral travel limit switch is triggered, the motor brake engages, and the vertical linear slide lowers the chair to the floor level, stopping at the lower limit switch (Figure 6.6). Once lowered, the chair is slid away from the system (Figure 6.7), and the foldable linkage is returned to its vertical position to free the staircase for normal use (Figure 6.8).

Following the design and simulation stages, experimental validation was carried out to assess the system's performance under practical operating conditions. Tests were conducted to verify the system's ability to maintain stable motion control and mechanical integrity during both ascent and descent. Key evaluation parameters, specifically chosen to address the limitations of existing systems, were applied to measure performance. A standard chair was used during testing to demonstrate the system's adaptability to conventional chair and wheelchair configurations (see Figure 7).

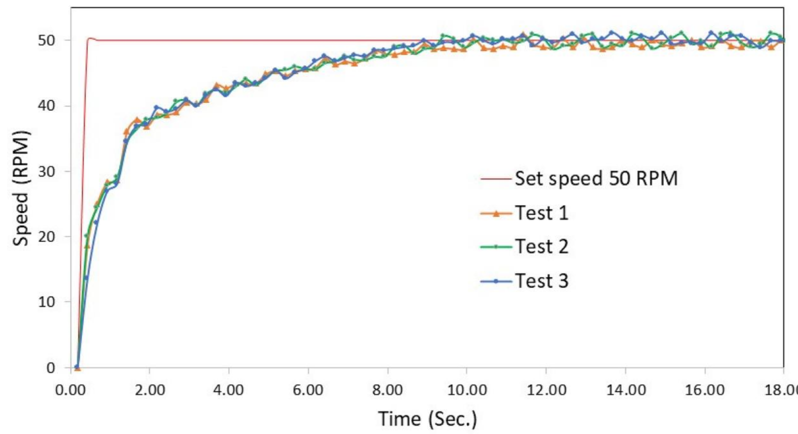

Figure 8. Repeatability validation: three consecutive ascending runs at 50 RPM on a 30° stair slope, confirming consistent system behavior across repeated trials.

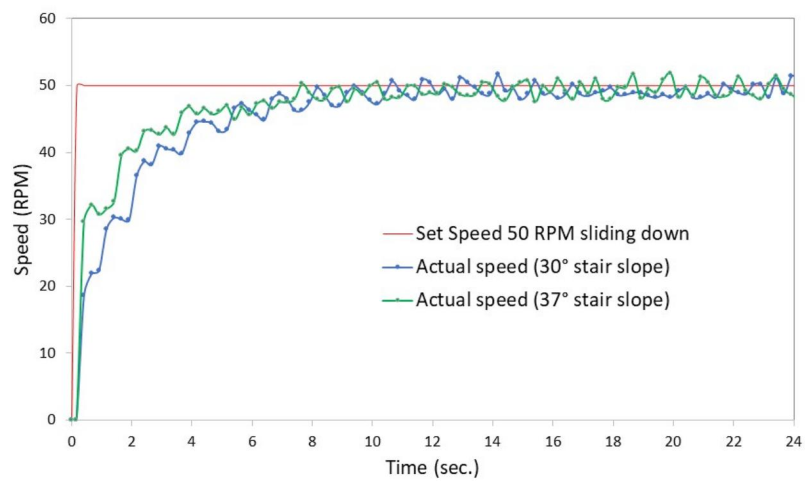

Figure 9. System performance at different stair slope angles: 30° and 37°.

4.1. Experimental validation

The validation process was designed to vary one parameter at a time while keeping all other conditions constant for each test. Each test condition was repeated three times to confirm repeatability. Figure 8 illustrates three repeated runs under identical conditions – ascending motion at 50 RPM on the same stair slope – demonstrating a high degree of consistency across trials. The plotted responses in Figures 9–12 represent characteristic runs selected from these repeated trials. Settling time was defined as the time elapsed from motion initiation until the velocity response remained within ±5% of the set-point value.

The conducted tests of Figures 9–12 included the following:

1. Slope variation: Stair angles between 30° and 37° were selected, as this range is widely recognized as optimal for residential and commercial stairs, offering a practical balance between comfort and safety. The tests were performed using a 5 kg load traveling down the stairs at 50 RPM speed.
2. Payload variation: Dead weights of 5 kg and 8 kg were used to simulate users of different masses. These tests involved ascending motion on a 30° stair slope at a 50 RPM travel speed.
3. Speed variation: The set travel speed was adjusted to assess the robustness and stability of the PID control system. Tests used a 5 kg mass ascending up a 30° stair slope angle at 60 and 70 RPM travel speed.
4. Direction of travel: Tests were performed during both ascent and descent to ensure consistent performance and reliability in either direction. A 5 kg mass traveled at 50 RPM travel speed up and down a 30° stair slope angle.

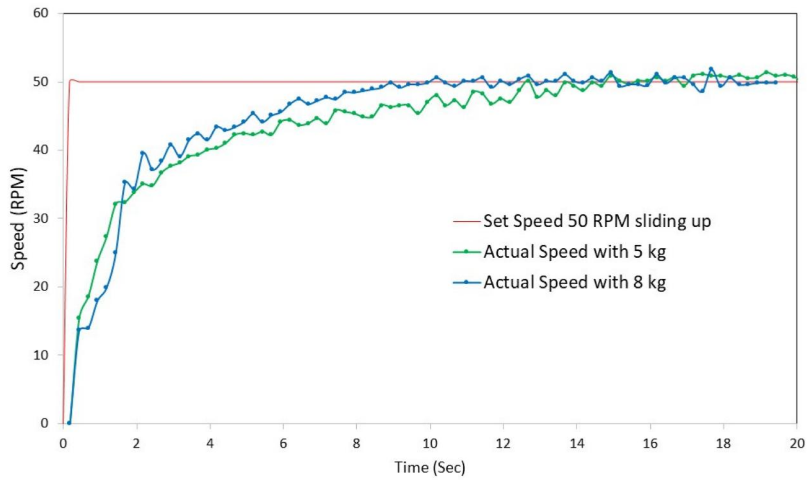

Figure 10. System performance for payloads of 5 kg and 8 kg.

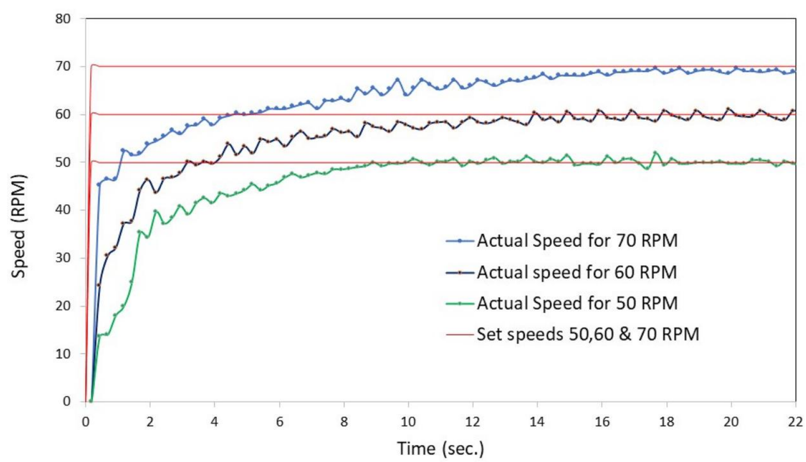

Figure 11. System response at different set speeds.

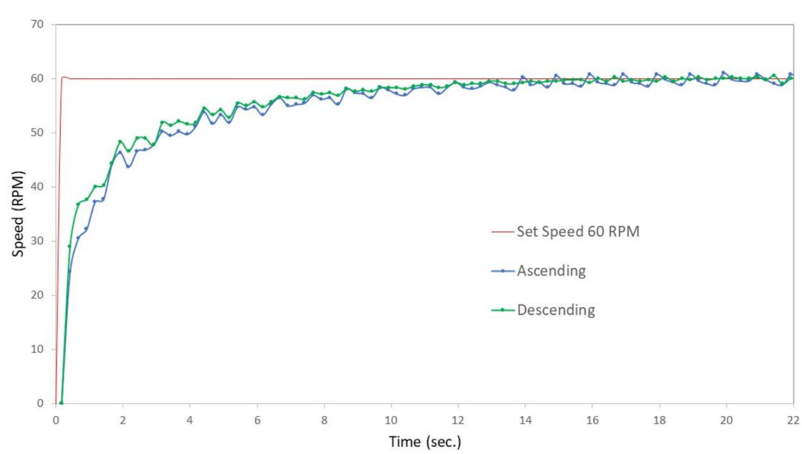

Figure 12. System response during ascending and descending motion.

The PID gains were tuned to achieve an overdamped response and sluggish transient response, preventing overshoot that could introduce abrupt jerks at the start of motion. This configuration ensures a smooth and gradual increase in velocity, enhancing user comfort. The system consistently demonstrated strong robustness in regulating travel speed across all operating conditions, effectively preventing over-speeding. Although the system remained overdamped with no overshoot in all tests, variations in the operating parameters influenced the settling time, causing the system to reach steady-state velocity at different rates, as explained later.

The resulting system responses and performance metrics for these test conditions are illustrated in Figures 9–12 (Yousef, 2025).

The effect of stair slope on system performance is illustrated in Figure 9. Increasing the stair angle while the user slides down resulted in a faster response, allowing the system to reach its set speed in a shorter time. While the set speed was achieved in approximately 10 seconds at a 30° slope, it was reached in about 6.8 seconds at the steeper angle.

Figure 10 shows the effect of varying the payload to simulate users of different weights while keeping all other parameters constant while sliding up the stairs. Although the proof-of-concept prototype was originally designed for a 5 kg payload, the validation tests were conducted using both 5 kg and 8 kg loads – representing more than a 60% increase over the rated mass. The system demonstrated strong robustness in velocity regulation, maintaining smooth and stable performance with an expected difference in settling time where Ts(5-kg)=8.4 s and Ts(8-kg) = 13.5 s.

Figure 11 presents the ascending-motion tests conducted at reference speeds of 50, 60, and 70 RPM. The system demonstrated stable and consistent performance, successfully settling at each target speed with an expected increase in settling time at higher reference speeds.

Figure 12 showed that the controller consistently managed to move the payload up and down the stairs, maintaining a constant velocity at the preset value of 50 RPM. The system, however, reached steady-state performance in a shorter time during descent. This was due to the gravitational force assisting the system's motion, which results in a faster transient response. This difference is evidenced by the settling time, which was 7.5 s during descent compared to 8.5 s during ascent.

4.2. Performance highlights

The system's ease of reconfigurability, highlighted in Table 2, underscores its expected practical advantages over existing alternatives. This feature is expected to enhance adaptability, usability, and long-term flexibility, addressing key limitations found in current implementations.

Overall, during both ascent and descent on the stairs, the system demonstrated that the velocity control mode is an effective option. This approach eliminates the need for prior knowledge about the user's weight or the type of chair, demonstrating smooth and controlled performance within the tested conditions without noticeable fluctuations in friction force or cable tension. Moreover, the system supports the accommodation of simple auxiliary devices, such as standard canes, which can be handled as an additional payload.

The following design features further contribute to the system's practical advantages:

- The system's operating speed, which determines lateral travel velocity, is user-defined and sets the velocity controller's target speed.
- Variations in stair step height and depth that affect the stair slope can be accommodated by adjusting the tilt angle of the guide rails while maintaining the vertical alignment of the lifting prismatic joint.
- The use of a rope-driven system guided by rails is designed to eliminate oscillatory motion, indicating a more stable and comfortable user experience in the prototype tests compared to systems that rely on tracked locomotion.
- Minimal modifications to existing staircases are required due to the system's simplified and modular design, suggesting ease of use, installation, maintenance, and removal.

Table 2. System's performance evaluation and design-based assessments.

Criterion	Performance
Payload capacity	Modular and customizable based on motor selection.
Operating speed	User-defined specified by the controller set-speed
Adaptability to various stair slopes/ step heights	The guide-rail slope angle can be adjusted during installation to accommodate various stair slopes and step dimensions.
Ride stability & comfort	Rope-driven mechanism with no oscillation due to sliding over smooth guide rails. This is anticipated based on rope-driven rail guidance and overdamped velocity control, pending human-subject validation.
Impact on existing staircase	Minimal modifications; foldable and aesthetically unobtrusive
Ease of installation & maintenance	Design-based assessment; simplicity of components and modular architecture suggest ease of installation and maintenance, pending field validation

4.3. Cost considerations

A key practical advantage of the proposed system is its simplified and modular design, which suggests lower expected cost and installation complexity relative to current assistive stair-navigation systems. While a full cost analysis is not yet feasible due to the use of a scaled-down prototype, a preliminary qualitative comparison can still be made. The primary components – two electric motors, guide rails, a cable-pulley system, and a microcontroller-based controller – are standard industrial parts that are generally inexpensive compared to the specialized actuators, multi-degree-of-freedom mechanisms, and advanced sensing units used in robotic stair-climbing wheelchairs. Installation is also straightforward, requiring only the attachment of guide rails to an existing staircase without structural modifications, unlike platform lifts or wheelchair elevators that require construction work, reinforcement, and regulatory approval. For reference, commercial stair lifts and platform lifts typically range from USD 5,000 to 15,000, excluding installation. In contrast, the proposed system is expected to be substantially less costly due to reduced mechanical complexity and the use of off-the-shelf components. Additionally, the modular architecture facilitates low-cost maintenance through simple replacement of motors, cables, or bearings. However, it should be noted that while the system's cost-effectiveness is argued qualitatively based on component simplicity, a rigorous quantitative cost analysis could not be conducted at the prototype stage.

4.4. Limitations

Several limitations of the current study should be acknowledged. First, all experimental tests were conducted using a scaled-down prototype with a maximum payload of 5 kg, and no human subject testing has been performed; therefore, claims regarding real-world usability, user comfort, and helper reduction remain engineering projections pending full-scale validation. Second, staircase geometry was limited to straight staircases; the performance of staircases on curved and spiral configurations has not been assessed. Finally, long-term durability, safety certification, and regulatory compliance testing are outside the scope of this work and will be essential prerequisites for any future commercial deployment.

4.5. Future works

Future work will focus on several key areas to enhance the system's functionality and applicability. While the current design primarily addresses standard staircase configurations, future research will focus on extending the system's adaptability to more complex staircase geometries, including curved staircases, spiral stairs, and irregular step designs. Also, enhancing the system's intelligence level by integrating environmental perception capabilities will be explored. This will involve incorporating sensors and algorithms to enable the system to perceive and adapt to its surroundings, moving beyond basic automated control to provide a more intuitive user experience. Moreover, although the present study emphasized modularity and evaluated performance using payloads appropriate for the scaled-down proof-of-concept prototype, future experiments will include testing with human subjects – pending the necessary ethical approvals, safety certification, and logistical preparations – to validate realistic performance and assess the system's suitability for commercial deployment, then a detailed cost analysis and installation procedures for residential and healthcare applications will be evaluated to further assess the system's overall commercial feasibility.

5. Conclusion

A practical stair-climbing assistive system for individuals with mobility impairments was proposed and evaluated as a proof-of-concept prototype. The system introduces several key innovations, including a piggyback-style mechanism with adaptable prismatic joints, automatic velocity control, and a foldable linkage for minimal staircase obstruction. Unlike conventional stair-climbing wheelchairs or track-mounted chairs, this system allows users to use standard chairs or any available wheelchair, showing potential to reduce reliance on external helpers in future full-scale implementations. Proof-of-concept experimental results confirmed smooth and stable navigation under controlled conditions. The system's features, such as its ability to adapt to varying stair inclinations and user payloads without requiring prior calibration, and its minimal impact on existing staircase structures, suggest its practicality and ease of use, pending further validation. Experimental validation confirms the feasibility of the system's operation, demonstrating smooth and comfortable movement with customizable parameters for diverse applications and various user scenarios. The foldable structure ensures that the stairs remain clear for other users when not in use, providing an aesthetically appealing design. The proof-of-concept results demonstrate promising design features – affordability, user-friendliness, and versatility – that merit further investigation toward real-world deployment in home and healthcare environments, subject to full-scale validation with human subjects. Future work will focus on enhancing the system's intelligence and expanding its adaptability to accommodate various staircase configurations.

Author contributions

CRediT: **Basem Yousef**: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing; **Omar Antar**: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft; **Yousef Abdelraouf**: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft; **Eissa Bu-Moza**: Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing – original draft; **Mohamad Kharseh**: Conceptualization, Investigation, Methodology, Visualization, Writing – review & editing; **Khaled Al-Aribe**: Conceptualization, Investigation, Methodology, Visualization, Writing – review & editing; **Khalifa Harib**: Conceptualization, Investigation, Methodology, Visualization, Writing – review & editing.

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Data availability statement

The data supporting the findings of this study are openly available in Figshare at the following DOI: https://doi.org/10.6084/m9.figshare.29631422.

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References

Alamdari, A., & Krovi, V. N. (2016). Design of articulated leg–wheel subsystem by kinetostatic optimization. *Mechanism and Machine Theory*, *100*, 222–234. https://doi.org/10.1016/j.mechmachtheory.2016.02.010

Bang, Y., Lee, C., Yoo, J., Lee, K., & Kim, I. (2011). Two-legged stair-climbing wheelchair and its stair dimension measurement using distance sensors. *11th International Conference on Control, Automation and Systems*, 1788–1791.

Chocoteco, J., Morales, R., & Feliu, V. (2015). Improving the climbing/descent performance of stair-climbing mobility systems confronting architectural barriers with geometric disturbances. *Mechatronics*, *30*, 11–26. https://doi.org/10.1016/j.mechatronics.2015.06.001

Ikeda, H., Toyama, T., Maki, D., Sato, K., & Nakano, E. (2021). Cooperative step-climbing strategy using an autonomous wheelchair and a robot. *Robotics and Autonomous Systems*, *135*, 103670. https://doi.org/10.1016/j.robot.2020.103670

Onozuka, Y., Akai, H., & Nakano, E. (2020 *Dynamic stability control of inverted-pendulum-type robotic wheelchair for going up and down stairs* [Paper presentation]. 2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 4114–4119. https://doi.org/10.1109/IROS45743.2020.9341242

Pappalettera, A., Bottiglione, F., Mantriota, G., & Reina, G. (2023). Watch the next step: A comprehensive survey of stair-climbing vehicles. *Robotics*, *12*(3), 74. https://doi.org/10.3390/robotics12030074

Quaglia, G., & Nisi, M. (2017). Design of a self-leveling cam mechanism for a stair climbing wheelchair. *Mechanism and Machine Theory*, *112*, 84–104. https://doi.org/10.1016/j.mechmachtheory.2017.02.003

Quaglia, G., Franco, W., & Oderio, R. (2011). Wheelchair.q, a motorized wheelchair with stair climbing ability. *Mechanism and Machine Theory*, *46*(11), 1601–1609. https://doi.org/10.1016/j.mechmachtheory.2011.07.005

Robert, B. (2019). Climbing robots: Recent research and emerging applications. *Industrial Robot*, *46*(6), 721–727.

Wei, C., Wu, J., Sun, J., Sun, H., Yao, Y., & Ruan, Q. (2022). Design of a passive locomotion closed-chain multi-legged platform for terrain adaptability. *Mechanism and Machine Theory*, *174*, 104936. https://doi.org/10.1016/j.mechmachtheory.2022.104936

Wen, H., Yang, H., Chen, Y., Zhou, L., & Wu, D. (2019). A robot with decoupled mechanical structure and adapted state machine control for both ground and staircase situations. *Applied Sciences*, *9*(23), 5185. https://doi.org/10.3390/app9235185

Yoneda, K., Ota, Y., & Hirose, S. (2009). High-grip stair climber with powder-filled belts. *The International Journal of Robotics Research*, *28*(1), 81–89. https://doi.org/10.1177/0278364908098091

Yousef, B. (2025). *PID velocity control data: Figs 8, 9, 10 & 11*. Figshare. https://doi.org/10.6084/m9.figshare.29631422.v3